Assignment 01

Networking Commands

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Commands:

1. ping — Test host reachability

SYNTAX

```
ping [options] destination
```

OPTIONS (6)

Option	Description
-c count	Send only a specific number of packets, then stop
-i interval	Wait this many seconds between sending each packet
-s size	Set the number of data bytes to send per packet
-t ttl	Set the IP Time-To-Live (max hops before packet is dropped)
-w deadline	Stop after this many seconds, regardless of packets sent
-q	Quiet mode — only show the summary at the end

EXAMPLE

```
ping -c 4 -i 1 8.8.8.8
```

```
user@user-HP-EliteBook-840-G1:~$ ping
ping: usage error: Destination address required
user@user-HP-EliteBook-840-G1:~$ ping -c 4 -i 1 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=247 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=322 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=307 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=692 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 246.888/392.021/692.209/175.583 ms
user@user-HP-EliteBook-840-G1:~$
```

2. traceroute — Trace the path to a host

SYNTAX

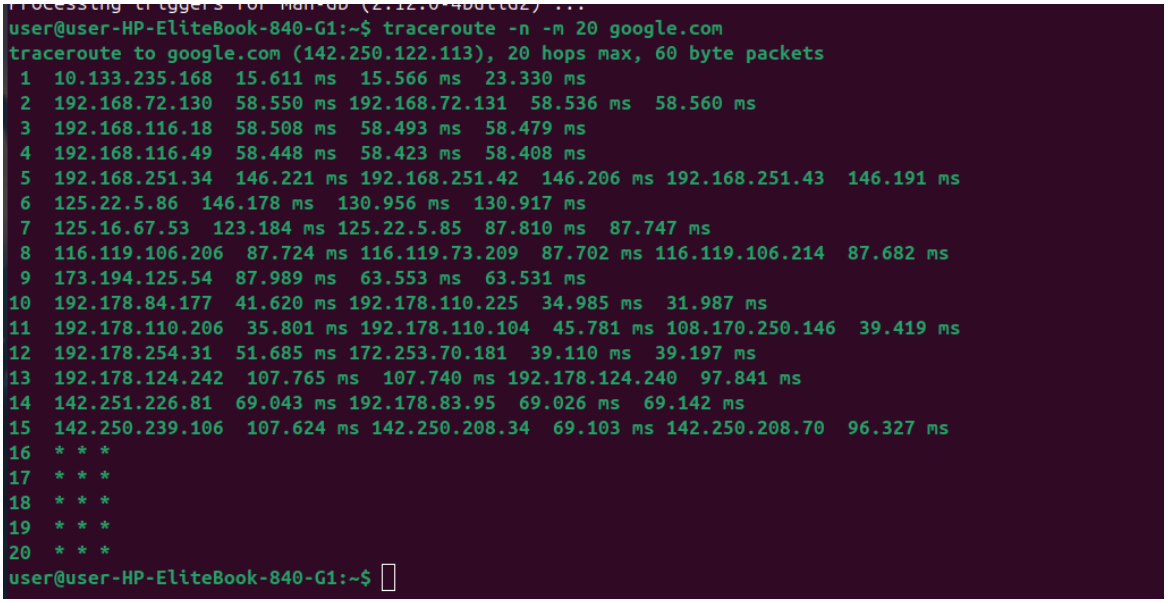
```
traceroute [options] destination
```

OPTIONS (6)

Option	Description
-m max_ttl	Set the maximum number of hops to probe
-n	Print hop addresses numerically, skip DNS resolution
-q nqueries	Number of probe packets sent per hop (default 3)
-w waittime	Seconds to wait for a response before timing out
-I	Use ICMP ECHO instead of UDP probes
-T	Use TCP SYN probes (useful when ICMP/UDP is blocked)

EXAMPLE

```
traceroute -n -m 20 google.com
```



3. ip — Show/configure interfaces & routes

SYNTAX

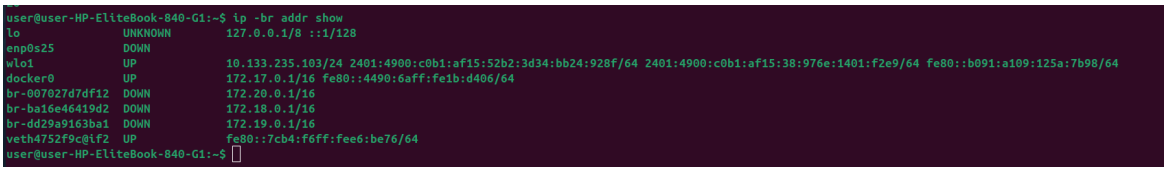
```
ip [options] object {show|add|del|set} ...
```

OPTIONS (6)

Option	Description
addr show	Display IP addresses assigned to all interfaces
link set dev up/down	Bring a network interface up or down
route show	Display the current routing table
-4 / -6	Restrict output to IPv4 or IPv6 only
-s	Show extra statistics (e.g. with link show)
-br	Show brief, condensed output

EXAMPLE

```
ip -br addr show
```



4. netstat — Show connections & listening ports

SYNTAX

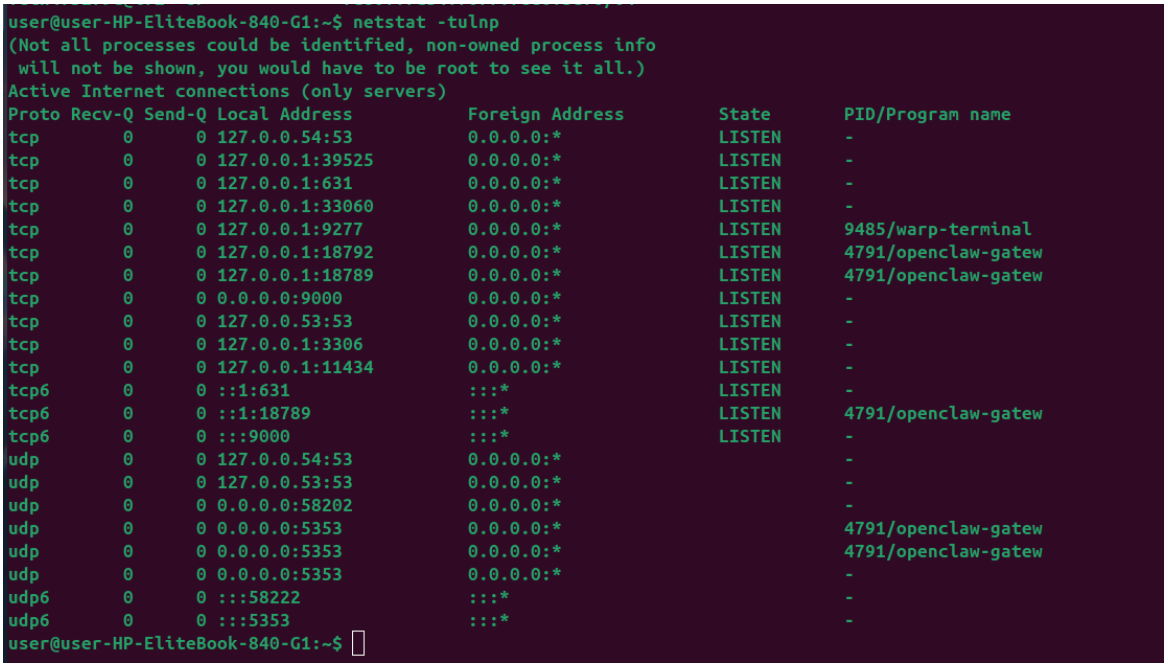
```
netstat [options]
```

OPTIONS (6)

Option	Description
-t	Show TCP connections
-u	Show UDP connections
-l	Show only listening sockets
-n	Show numeric addresses/ports instead of resolving names
-p	Show the PID and program name using each socket
-r	Display the kernel routing table

EXAMPLE

```
netstat -tulnp
```



5. ss — Modern socket statistics

SYNTAX

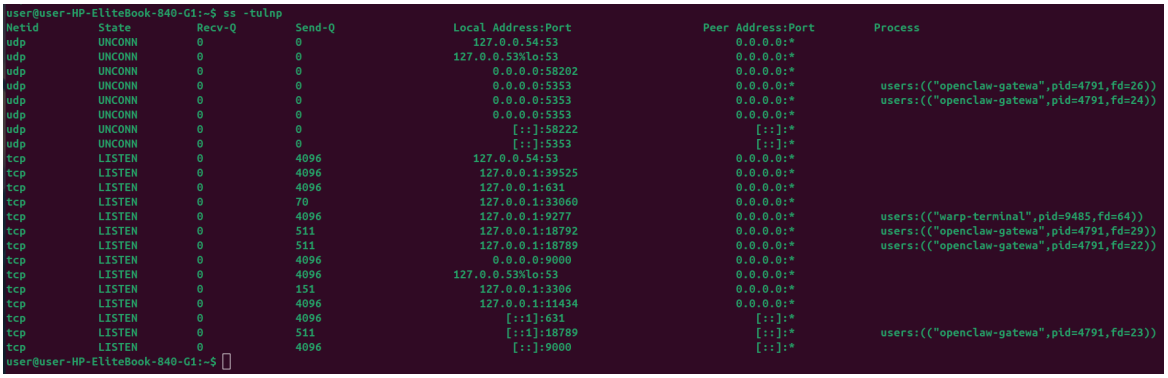
```
ss [options]
```

OPTIONS (6)

Option	Description
-t	Show TCP sockets
-u	Show UDP sockets
-l	Show only listening sockets
-n	Don't resolve service names, show numeric ports
-p	Show the process using each socket
-s	Print summary statistics

EXAMPLE

```
ss -tulnp
```



6. nslookup — Query DNS records

SYNTAX

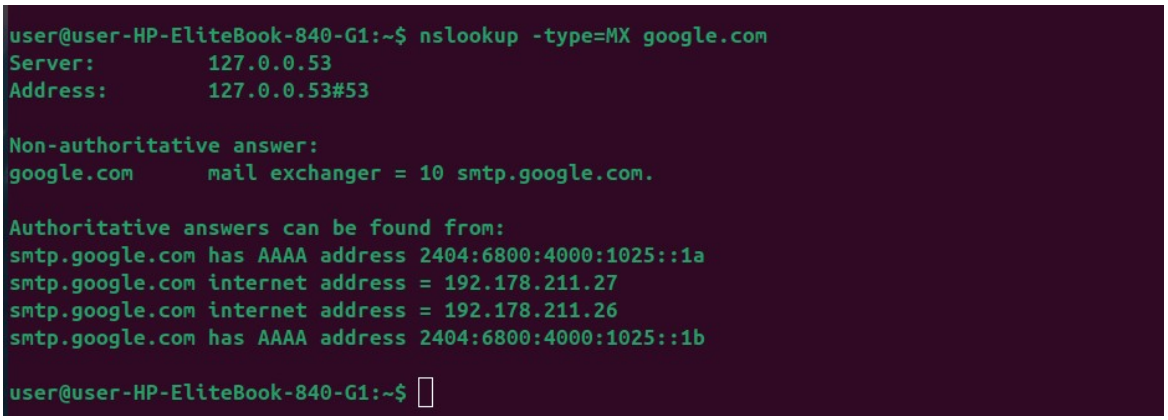
```
nslookup [options] host [server]
```

OPTIONS (6)

Option	Description
-type=A|MX|NS|TXT	Specify which DNS record type to query
-debug	Show detailed query/response information
-timeout=n	Set the timeout (seconds) for each request
-retry=n	Set the number of retries on failure
-port=n	Query a DNS server on a non-standard port
server	Specify which DNS server to query (last argument)

EXAMPLE

```
nslookup -type=MX google.com 8.8.8.8
```



7. dig — Detailed DNS lookup

SYNTAX

```
dig [@server] [options] name [type]
```

OPTIONS (6)

Option	Description
-x	Perform a reverse DNS lookup (IP to hostname)
+short	Print only the short answer, no extra output
+trace	Trace the resolution path from root DNS servers down
-t type	Specify the DNS record type (A, MX, TXT, etc.)
-p port	Query a DNS server on a specific port
@server	Send the query to a specific DNS server

EXAMPLE

```
dig @8.8.8.8 +short google.com MX
```

```
user@user-HP-EliteBook-840-G1:~$ nslookup -type=MX google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
google.com      mail exchanger = 10 smtp.google.com.

Authoritative answers can be found from:
smtp.google.com has AAAA address 2404:6800:4000:1025::1a
smtp.google.com internet address = 192.178.211.27
smtp.google.com internet address = 192.178.211.26
smtp.google.com has AAAA address 2404:6800:4000:1025::1b

user@user-HP-EliteBook-840-G1:~$ dig @8.8.8.8 +short google.com MX
10 smtp.google.com.
user@user-HP-EliteBook-840-G1:~$ dig @8.8.8.8  google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.5-Ubuntu <<>> @8.8.8.8 google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 51794
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             187     IN      A       192.178.211.138
google.com.             187     IN      A       192.178.211.102
google.com.             187     IN      A       192.178.211.100
google.com.             187     IN      A       192.178.211.101
google.com.             187     IN      A       192.178.211.113
google.com.             187     IN      A       192.178.211.139

;; Query time: 33 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Thu Jul 23 08:54:29 IST 2026
;; MSG SIZE  rcvd: 135

user@user-HP-EliteBook-840-G1:~$
```

8. ssh — Secure remote shell

SYNTAX

```
ssh [options] user@host
```

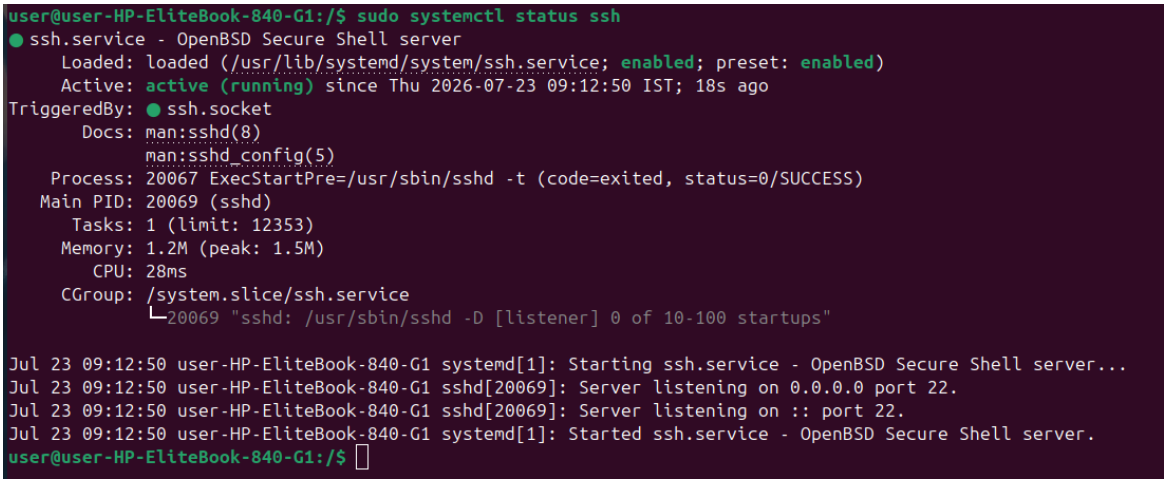
OPTIONS (6)

Option	Description
-p port	Connect to a non-default SSH port
-i identity_file	Use a specific private key for authentication
-L local:host:remote	Forward a local port to a remote address
-R remote:host:local	Forward a remote port back to the local machine

-D port	Create a dynamic SOCKS proxy through the SSH tunnel
-v	Verbose mode, useful for debugging connection issues

EXAMPLE

```
sudo systemctl status ssh
```



9. scp — Secure copy over SSH

SYNTAX

```
scp [options] source destination
```

OPTIONS (6)

Option	Description
-P port	Connect using a non-default SSH port (capital P)
-r	Recursively copy entire directories
-i identity_file	Use a specific private key for authentication
-C	Compress data during transfer
-p	Preserve file modification times and permissions
-v	Verbose mode for troubleshooting transfers

EXAMPLE

```
scp -P 2222 -r ./project user@host:/var/www/
```

10. rsync — Efficient file synchronization

SYNTAX

```
rsync [options] source destination
```

OPTIONS (6)

Option	Description
-a	Archive mode — preserves permissions, timestamps, symlinks

-v	Verbose output showing files being transferred
-z	Compress file data during the transfer
--delete	Delete files at destination that no longer exist at source
-e ssh	Specify the remote shell to use (commonly SSH)
-r	Recurse into directories

EXAMPLE

```
rsync -avz --delete ./site/ user@host:/var/www/site/
```

11. curl — Transfer data with URLs

SYNTAX

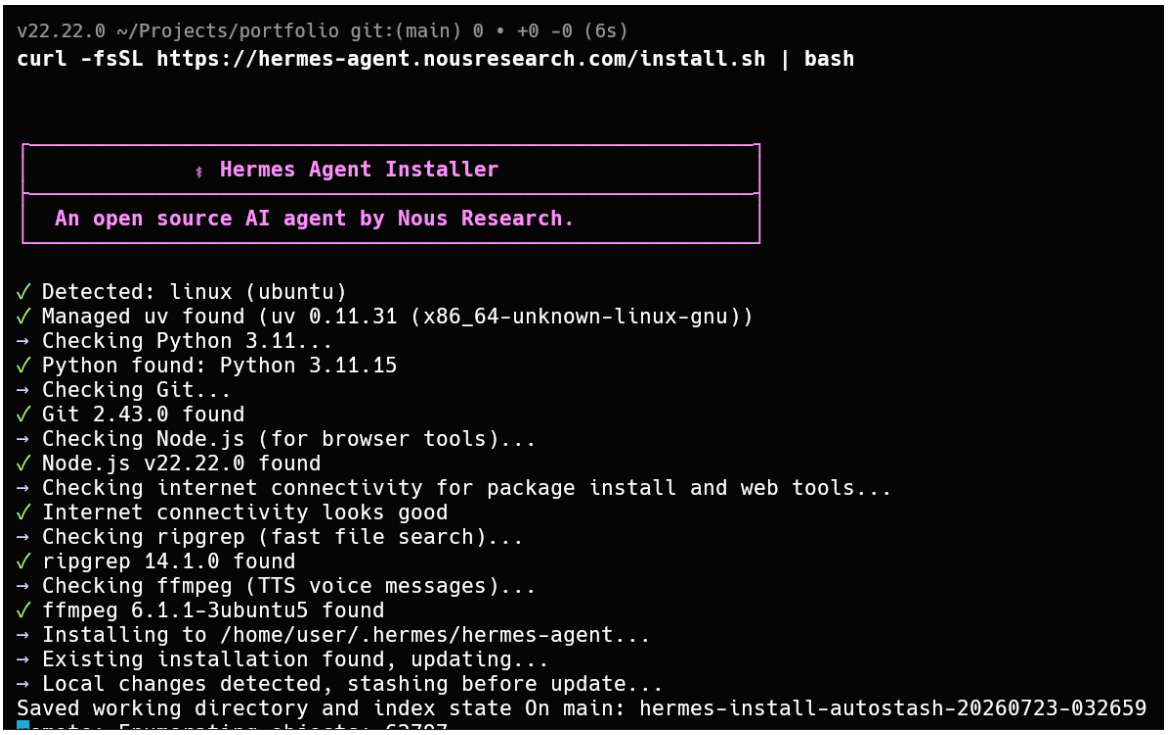
```
curl [options] URL
```

OPTIONS (6)

Option	Description
-O	Save the output to a file, using the remote filename
-I	Fetch headers only (HEAD request)
-X method	Specify the HTTP request method (GET, POST, etc.)
-H header	Add a custom request header
-d data	Send data in the request body (POST by default)
-L	Follow HTTP redirects automatically

EXAMPLE

```
curl -X POST -H "Content-Type: application/json" -d '{"a":1}'
https://api.example.com
```



12. wget — Download files from the web

SYNTAX

```
wget [options] URL
```

OPTIONS (6)

Option	Description
-r	Download recursively (e.g. entire site sections)
-np	Don't ascend to the parent directory when recursing
-c	Continue/resume a partially downloaded file
-q	Quiet mode, suppress output
-O file	Save output with a specific filename
--limit-rate=rate	Limit the download speed (e.g. 200k)

EXAMPLE

```
wget -c --limit-rate=500k -O file.zip https://example.com/file.zip
```

13. nmap — Network/port scanner

SYNTAX

```
nmap [options] target
```

OPTIONS (6)

Option	Description
-p range	Scan a specific port or port range
-sV	Detect service/version info on open ports

-sS	Perform a stealthy TCP SYN scan
-O	Attempt operating system detection
-sn	Ping scan only — discover live hosts, skip port scan
-A	Aggressive scan: OS, version, script, and traceroute detection

EXAMPLE

```
nmap -sV -p 1-1000 192.168.1.0/24
```

14. tcpdump — Capture and inspect packets

SYNTAX

```
tcpdump [options] [expression]
```

OPTIONS (6)

Option	Description
-i interface	Capture packets on the specified interface
-w file	Write captured packets to a file
-r file	Read and display packets from a saved capture file
-n	Don't resolve hostnames/ports, show numeric values
-c count	Stop after capturing this many packets
-v	Verbose output with more packet detail

EXAMPLE

```
tcpdump -i eth0 -n port 443 -c 50
```

15. iptables — Configure Linux firewall rules

SYNTAX

```
iptables [options] -A|-D|-I chain rule-spec
```

OPTIONS (6)

Option	Description
-L	List current firewall rules
-A chain	Append a new rule to the end of a chain
-D chain	Delete a matching rule from a chain
-F	Flush (remove) all rules in a chain
-P chain policy	Set the default policy for a chain (ACCEPT/DROP)
-I chain	Insert a rule at a specific position in a chain

EXAMPLE

```
iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

16. ufw — Simplified firewall management

SYNTAX

```
ufw [options] command
```

OPTIONS (6)

Option	Description
allow port/proto	Allow traffic on a port or service
deny port/proto	Block traffic on a port or service
status	Show current firewall status and active rules
enable / disable	Turn the firewall on or off
delete rule	Remove an existing rule
reload	Reload the firewall to apply rule changes

EXAMPLE

```
ufw allow 443/tcp && ufw enable
```

17. arp — View/manage the ARP cache

SYNTAX

```
arp [options] [hostname]
```

OPTIONS (6)

Option	Description
-a	Display all current ARP table entries
-d hostname	Delete an entry from the ARP table
-s hostname mac	Add a static ARP entry
-n	Show numeric IPs, skip hostname resolution
-v	Verbose output
-i interface	Limit the operation to a specific interface

EXAMPLE

```
arp -a -n
```

```
user@user-HP-EliteBook-840-G1:/$ arp -a -n
? (10.133.235.168) at 42:44:30:43:04:0e [ether] on wlo1
user@user-HP-EliteBook-840-G1:/$
```

18. ip route — Manage the routing table

SYNTAX

```
ip route {show|add|del|get|replace} ...
```

OPTIONS (6)

Option	Description
show	Display the full routing table
add dest via gw	Add a new route to a destination via a gateway
del dest	Remove a route from the table
get dest	Show which route/interface would be used for a destination
replace dest via gw	Replace an existing route atomically
show table main	Show routes in a specific routing table

EXAMPLE

```
ip route add 10.0.0.0/24 via 192.168.1.1
```

19. iperf3 — Measure network throughput

SYNTAX

```
iperf3 [options]
```

OPTIONS (6)

Option	Description
-s	Run in server mode, waiting for a client to connect
-c host	Run in client mode, connecting to the given server
-p port	Use a specific port instead of the default (5201)
-t seconds	Set the test duration in seconds
-u	Use UDP instead of the default TCP
-P n	Run n parallel streams for higher aggregate throughput

EXAMPLE

```
iperf3 -c 10.0.0.5 -t 30 -P 4
```

20. nc (netcat) — Read/write over network connections

SYNTAX

```
nc [options] host port
```

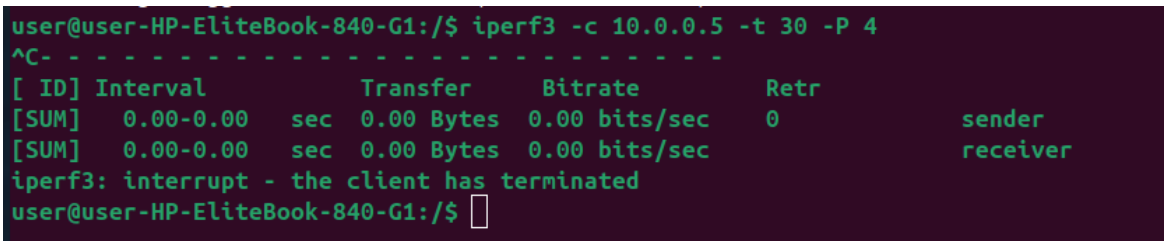
OPTIONS (6)

Option	Description
-l	Listen for an incoming connection instead of connecting out
-p port	Specify the local port to use
-v	Verbose mode, prints connection details
-z	Zero-I/O scanning mode — just test if a port is open

-u	Use UDP instead of the default TCP
-w seconds	Set a connection/idle timeout

EXAMPLE

```
nc -zv 192.168.1.10 20-100
```



Conclusion:

The program successfully Networking commands , Understand Networking Commands and implantation of those commands.